MARLON A. RIVERA

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Engineering Portfolio:
https://marlonrivera-automationengineer.github.io/

Dear Hiring Professionals,

Subject: Application for Senior Automation Design & Controls Engineer

Dear Hiring Manager,

I am writing to express my interest in joining your organization as a **Senior Automation Design & Controls Engineer**. With almost **14 years of international experience** in industrial automation, machine design, PLC/HMI programming, robotics integration, Smart Manufacturing, and continuous improvement, I am excited about the opportunity to contribute my engineering expertise to your manufacturing organization.

Throughout my career, I have successfully designed, developed, programmed, commissioned, and improved automated manufacturing systems across the **Philippines, Thailand, Vietnam, Korea, and Cambodia**. My experience covers the complete project lifecycle—from concept development and mechanical design through PLC programming, electrical integration, machine commissioning, production validation, and continuous improvement.

Currently, I serve as the **Assistant Manager – Automation & Maintenance** at **NIDEC SC WADO (Cambodia) Co., Ltd.**, where I lead automation and maintenance initiatives supporting large-scale manufacturing operations. Working closely with executive management, I oversee automation projects, machine reliability improvement, Smart Manufacturing implementation, production monitoring, maintenance strategy, and engineering activities focused on increasing productivity while reducing equipment downtime and operating costs.

During my career, I have delivered automation solutions including automated laser welding systems, robotic production cells, automatic visual inspection systems, Smart Manufacturing monitoring systems, automated diecasting production lines, automated cleaning systems, AGV concepts, and machine transfer automation. These projects have improved manufacturing efficiency, product quality, production capacity, and operational reliability while reducing manual processes.

My technical expertise includes:

- Industrial Automation & Controls Engineering
- PLC Programming (Mitsubishi, Panasonic, Omron, Keyence, LSIS)
- HMI Development
- Robotics Integration (ABB, Nachi, Yamaha)
- Servo Motion Control
- Machine Design (SolidWorks & AutoCAD)
- Vision Inspection Systems
- Pneumatic & Industrial Automation Systems
- Smart Manufacturing & Industry 4.0
- OEE Improvement & Production Monitoring
- Factory Acceptance Testing (FAT)
- Site Acceptance Testing (SAT)

Beyond technical capability, I bring leadership experience in managing multidisciplinary engineering teams, collaborating with international customers and suppliers, mentoring engineers, and successfully delivering projects from concept through production handover. I believe that successful automation is not only about technology, but also about building practical solutions that improve safety, quality, productivity, and long-term manufacturing performance.

I am seeking an opportunity to contribute my engineering knowledge, leadership experience, and passion for industrial automation within a progressive manufacturing organization that values innovation, operational excellence, and continuous improvement. I am confident that my international manufacturing background and hands-on engineering experience would allow me to contribute immediately while continuing to grow professionally.

As a Philippine citizen currently employed in Cambodia, I am available for international relocation and would require employer-sponsored work authorization where applicable. I am fully prepared to support the relocation and visa process and am available to travel or relocate according to business requirements.

Thank you for considering my application. I would welcome the opportunity to discuss how my engineering experience and leadership background can contribute to your organization. I appreciate your time and consideration and look forward to the opportunity to speak with you.

Yours sincerely,

Mr. Marlon A. Rivera

MARLON A. RIVERA

Senior Automation & Controls Engineer

Machine Design | Automation Programmer | Robotics | Smart Manufacturing

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PROFESSIONAL PROFILE

Results-driven Senior Automation and Controls Engineer with 13+ years of experience in industrial automation, machine design, PLC/HMI programming, robotics integration, servo motion control, vision systems, pneumatic systems, and smart factory implementation across the Philippines, Vietnam, Korea, Thailand, and Cambodia. Experienced in delivering complete automation projects from concept design, mechanical design, assembly, wiring, programming, installation, qualification, commissioning, and production handover.

Proven track record of leading cross-functional engineering teams, delivering turnkey automation projects, improving Overall Equipment Effectiveness (OEE), reducing machine downtime, optimizing production cycle time, and implementing Industry 4.0 solutions that improve manufacturing productivity and quality. Hands-on experience with Mitsubishi, Siemens, Omron, Keyence, Schneider, Panasonic, Festo, Pro-face, ABB, Nachi, Yamaha, IAI, THK, SMC, SolidWorks, AutoCAD, Python, and factory monitoring systems.

Recognized for designing and programming high-impact automation systems including Auto Laser Weld, AVMI visual inspection automation, Auto Spray Machine, robotic deburring systems, diecast automation, AGV systems, and production monitoring solutions across the Philippines, Thailand, Cambodia, Vietnam, and Korea.

KEY CAREER HIGHLIGHTS

• 13+ years of international experience in Industrial Automation, Machine Design, and Smart Manufacturing.
• Successfully delivered automation projects across the Philippines, Thailand, Vietnam, Korea, and Cambodia.
• Designed, programmed, and commissioned industrial automation systems including robotic cells, vision inspection systems, laser welding machines, AGV concepts, Smart Factory monitoring systems, and automated production lines.
• Experienced in leading multidisciplinary engineering teams, managing automation projects from concept design through commissioning and production handover.
• Recipient of the Employee of the Year Award (2016) and recognized through a Korean Government Patent for engineering innovation.
• Strong expertise in PLC Programming, Robotics Integration, Servo Motion Control, Vision Systems, OEE Improvement, Industry 4.0, and Continuous Improvement.

CORE COMPETENCIES

Industrial Automation | PLC Programming | HMI Development | Machine Design | Controls Engineering | Robotics Integration | Servo Motion Control | Vision Systems | Pneumatic Systems | Smart Factory | OEE Improvement | Cycle Time Reduction | Production Troubleshooting | Preventive Maintenance | Machine Commissioning | FAT / SAT | Machine Qualification | CPK / GRR / Correlation | Root Cause Analysis | Continuous Improvement | Lean Manufacturing | Project Leadership | Vendor Coordination | Team Leadership

TECHNICAL SKILLS MATRIX

Industrial Automation & Controls	PLC Programming (Mitsubishi GX Works2/GX Works3, Siemens TIA Portal, Omron CX-One/Sysmac, Keyence KV Studio) HMI Development (Mitsubishi GOT, Pro-face, Beijer, Panasonic, Weintek) Servo Motion Control, Motion Programming, Industrial Networking Machine Commissioning, FAT / SAT, Machine Qualification, Production Troubleshooting
Robotics & Motion Systems	ABB Industrial Robots, Nachi Robots, Yamaha Robots, IAI Actuators (ASEL / PSEL), THK Motion Systems SMC Pneumatic Systems, Servo Motors, Stepper Motors, Multi-Axis Motion Systems
Mechanical Design	SolidWorks (3D/2D), AutoCAD, Machine Design, Mechanical Assembly Design Production Fixtures, Tooling Design, Conveyor Systems, Safety Machine Design
Smart Factory & Software	Python, FastAPI, Manufacturing Monitoring Systems, OEE Monitoring, Production Dashboard Development Factory Data Collection, Cycle Time Analysis, SQL (Basic), Microsoft Office
Vision & Inspection Systems	Keyence Vision Systems, Auto Visual Inspection, Camera Inspection Systems, Sensor Integration, Barcode Systems
Manufacturing Engineering	Lean Manufacturing, Continuous Improvement, Root Cause Analysis, OEE Improvement Preventive Maintenance, Predictive Maintenance, Process Optimization, CAPA, CPK, GRR, Correlation
Leadership	Engineering Team Leadership, Project Management, Customer Technical Support, Vendor Coordination Cost Reduction, Technical Documentation, Engineering Training, Cross-functional Collaboration

PROFESSIONAL EXPERIENCE

Assistant Manager - Automation & Maintenance | NIDEC SC WADO Cambodia

Poipet, Cambodia | Jul 2018 – Present

Promoted into a leadership role responsible for driving factory automation, machine design, maintenance strategy, and Smart Factory initiatives for a global precision manufacturing company. Lead multidisciplinary engineering teams while reporting directly to executive management and supporting continuous manufacturing improvement across multiple production departments.

Key Responsibilities & Achievements

- Lead Automation & Maintenance Department supporting over 100 production lines and 60 engineers/technicians, managing engineering activities, manpower planning, machine reliability, and continuous improvement initiatives.
- Report directly to executive management (COO and Directors) regarding automation strategies, maintenance performance, production improvements, project progress, and capital investment planning.

- Design, develop, and commission fully automated manufacturing equipment from concept through production handover, including mechanical design, electrical integration, PLC programming, HMI development, assembly, testing, validation, and commissioning.
- Develop industrial automation solutions using Mitsubishi PLCs, HMIs, servo systems, industrial robots, pneumatic systems, sensors, safety devices, and industrial communication networks.
- Developed Smart Factory monitoring systems supporting real-time monitoring across more than 1,000 production machines by developing production monitoring systems, machine monitoring dashboards, OEE analysis tools, production data collection systems, and real-time manufacturing visualization.
- Analyze production bottlenecks, machine downtime, Overall Equipment Effectiveness (OEE), cycle time, and equipment utilization to identify improvement opportunities and increase manufacturing productivity.
- Lead cross-functional engineering projects involving Production, Quality, Maintenance, Manufacturing Engineering, and Management teams to ensure successful project execution.
- Perform root cause analysis for complex equipment failures and implement long-term corrective actions to improve machine reliability and reduce production downtime.
- Coordinate equipment installation, machine qualification, Factory Acceptance Testing (FAT), Site Acceptance Testing (SAT), production validation, and operator training.
- Support cost-reduction initiatives by improving automation efficiency, reducing manual labor, minimizing machine downtime, and optimizing manufacturing processes.
- Collaborate with international suppliers, equipment manufacturers, and technical partners throughout machine development, equipment procurement, installation, and commissioning.

Automation Engineer | Nissho Seiko Thailand

Thailand | Aug 2017 - Jul 2018

Served as the primary Automation Engineer responsible for delivering complete automation solutions from concept through production. Worked independently on mechanical design, electrical integration, PLC programming, HMI development, machine assembly, installation, commissioning, and production support for high-volume manufacturing.

Key Responsibilities & Achievements

- Led the Automation Section and reported directly to the Engineering Manager.
- Designed and developed fully automated manufacturing equipment to improve production efficiency, product quality, and process consistency.
- Integrated Mitsubishi PLCs, industrial sensors, pneumatic systems, servo motors, and HMI systems into production equipment.
- Developed automation improvements that reduced manual operations and increased manufacturing productivity.
- Performed machine troubleshooting, preventive maintenance, and process optimization to minimize equipment downtime.
- Supported production engineering teams by resolving automation-related issues and implementing long-term engineering improvements.
- Designed and implemented an Automatic Visual Inspection (AVI) system that improved final inspection consistency and reduced dependence on manual inspection.
- Worked closely with production, quality, and maintenance departments to ensure smooth equipment handover and stable production performance.

Research and Development Supervisor | JIO MHW Manufacturing Corporation (JMM)

Philippines | Jul 2015 - Apr 2017

Promoted to lead the Design and Programming Section, managing engineering projects for international customers across the Philippines, Vietnam, and Korea. Responsible for transforming customer requirements into complete automation solutions, leading multidisciplinary engineering teams, and delivering innovative manufacturing equipment from concept through commissioning.

Key Responsibilities & Achievements

- Led the Research & Development Design and Programming Section, supervising engineers and assigning project responsibilities while reporting directly to the Engineering Director.
- Worked directly with local and international customers to understand manufacturing requirements, evaluate technical feasibility, and develop complete automation concepts.
- Managed multiple automation projects simultaneously, coordinating project schedules, engineering resources, manufacturing activities, and customer technical reviews.
- Designed complete automated manufacturing systems using SolidWorks and AutoCAD, including mechanical structures, production fixtures, pneumatic systems, safety mechanisms, and machine layouts.
- Developed PLC and HMI control systems for fully automated production equipment, integrating sensors, servo systems, pneumatic devices, and industrial automation components.
- Successfully designed and delivered the Auto Spray Machine for Samsung Camera production, which was later upgraded into Version 2 and supplied to IM Vietnam.
- Led the development of the Auto Air Blow Machine for Dae Hyung / LG Korea, improving production quality and manufacturing consistency.
- Received recognition associated with the Auto Spray Machine project through a Korean Government patent, demonstrating innovation in industrial automation and machine design.
- Collaborated closely with fabrication, machining, electrical assembly, purchasing, and production departments to ensure projects were completed on schedule and met customer specifications.
- Conducted machine testing, customer demonstrations, Factory Acceptance Testing (FAT), troubleshooting, installation support, and production validation.

- Prepared engineering documentation including machine manuals, design drawings, bills of materials (BOM), pneumatic diagrams, electrical layouts, and technical reports.
- Provided technical guidance and mentoring to junior engineers, programmers, and designers to improve engineering quality and project delivery.

Major Accomplishments

- Designed multiple automation systems successfully deployed in the Philippines, Vietnam, and Korea.
- Successfully delivered the internationally recognized Auto Spray Machine platform for Samsung Camera production.
- Received Employee of the Year – 2016 in recognition of outstanding engineering performance, innovation, leadership, and contribution to the company's success.

Technician II - Instrumentation / Automation | NIDEC Philippines Corporation (NCFL)

Philippines | May 2013 - Jun 2015

Built a strong engineering foundation in industrial automation by developing PLC-controlled production equipment, robotics applications, machine vision solutions, and manufacturing process improvements for high-volume precision manufacturing. Progressed into a technical leadership role supporting production, engineering, and automation initiatives.

Key Responsibilities & Achievements

- Designed, developed, and programmed industrial automation equipment using PLCs, HMIs, industrial sensors, servo systems, and 3-axis robotic systems.
- Participated in complete machine development activities including concept generation, mechanical improvement, PLC programming, HMI development, electrical troubleshooting, testing, and production support.
- Developed automation projects that improved production efficiency, product quality, and manufacturing consistency while reducing manual operations.
- Designed and programmed the Auto Laser Weld System for Apple product manufacturing, supporting production capacity of approximately 50,000 units per production line per month.
- Performed production equipment troubleshooting involving PLC controls, sensors, pneumatic systems, servo drives, electrical systems, and industrial automation devices.
- Supported machine qualification activities including CPK, GRR, Correlation Studies, production validation, and engineering verification.
- Worked closely with Manufacturing Engineering, Production, Quality Assurance, and Maintenance departments to resolve complex technical issues and improve equipment performance.
- Assisted in continuous improvement initiatives focused on reducing machine downtime, increasing production output, improving equipment reliability, and minimizing manufacturing defects.

Major Accomplishments

- Successfully developed the Auto Laser Weld System supporting high-volume Apple product manufacturing.
- Eliminated manual visual inspection through the successful implementation of the Automatic Visual Machine Inspection (AVMI) system.
- Demonstrated strong technical capability in automation, robotics, machine qualification, and production support, leading to career advancement into supervisory and management positions.

SELECTED AUTOMATION PROJECTS & IMPACT

AUTO LASER WELDING SYSTEM (Apple Products) | Philippines:

Designed and commissioned a fully automated laser welding system for Apple product manufacturing supporting approximately 50,000 units per production line per month, improving welding precision, production consistency, and manufacturing throughput.

Technologies

- PLC Programming | YAG Laser | Servo Motion | Precise Automation

AUTO SPRAY MACHINE | Philippines:

Led the complete design and automation development of a fully automatic spray machine for Samsung Camera production. The project was later enhanced into Version 2 and supplied internationally to IM Vietnam. The innovation received recognition through a Korean Government patent.

Technologies

- SolidWorks | PLC Programming | Pneumatics | Servo Systems | Industrial Automation

AUTO VACCUUM AND AIR CLEANING MACHINE | Korea:

Designed and developed an automatic air blow and vacuum system for Dae Hyung / LG Korea to improve production quality, process consistency, and manufacturing efficiency.

Technologies

- PLC, Pneumatic Systems | Mechanical Design | Automation Controls

6 BATH AUTOMATIC CLEANING MACHINE | Vietnam:

Designed and commissioned a fully automated six-bath production system for IM Vietnam, integrating mechanical automation, PLC controls, pneumatic systems, and production handling to improve process efficiency and product quality.

SMART FACTORY MONITORING SYSTEM | Cambodia:

Designed and programmed real-time manufacturing monitoring systems for production analysis, equipment monitoring, and continuous improvement. Developed OEE dashboards and production visualization tools to support Smart Factory initiatives.

Technologies

- Python |FastAPI | PLC Integration | Manufacturing Dashboards | OEE Monitoring

FULL AUTOMATIED DIECASTING PRODUCTION LINE | Cambodia:

Designed and implemented a fully automated diecast production system integrating transfer automation, robotic handling, servo motion, and machine synchronization to reduce manual operations and increase production efficiency.

Technologies

- PLC Integration | ABB (6 Axis Robot) | LCPL (4 Axis Robot) | Laser Marking (Keyence and Panasonic) | Hydraulic Auto-breaking machine

ROBOTIC DEBURRING+KNOCKING AUTOMATION CELL | Cambodia:

Designed robotic deburring systems using industrial robots to improve product consistency, reduce manual labor, and increase manufacturing productivity.

Technologies

- PLC Integration | Nachi (6 Axis Robot) | LCPL | Keyence Probe Sensors | Omron Servo Motor

AGV & PRODUCTION TRANSFER AUTOMATION | Cambodia:

Designed automation concepts supporting automated material transfer and AGV-based production systems to improve manufacturing logistics and production flow.

FULLY AUTOMATIC VISUAL INSPECTION SYSTEM (AVMI) | Cambodia:

Designed fully automated visual inspection equipment integrated into the final production process, improving inspection reliability while minimizing operator intervention.

Technologies

- Keyence (CCD Camera) | Keyence (Barcode Reader Camera) | Visual and Measurement Checking

MACHINE DESIGN & AUTOMATION PORTFOLIO:

Successfully designed, programmed, commissioned, and supported multiple industrial automation systems across the Philippines, Thailand, Vietnam, Korea, and Cambodia, covering:

- Machine Design, PLC Programming, HMI Development, Robotics Integration, Servo Motion Control
- Vision Systems, Pneumatic Systems, Smart Factory, Machine Commissioning, Continuous Improvement

AWARDS & RECOGNITIONS

Employee of the Year: JIO MHW Global Channel Manufacturing Corporation | 2016

Awarded Employee of the Year in recognition of outstanding engineering performance, innovation, leadership, and contribution to industrial automation projects.

Korean Government Patent Recognition| 2016

Recognized for engineering innovation through the successful development of the Auto Spray Machine, which received patent recognition from the Korean Government.

PROFESSIONAL TRAINING & CERTIFICATIONS

SolidWorks Essentials Course: DETI – Authorized SolidWorks Training Center | 2019

Topics Covered:

SolidWorks Part Modeling, Assembly Design, Engineering Drawings, and Mechanical Design Best Practices

Mechanical Design Training : JIO MHW Global Channel Manufacturing Corporation | 2016

Training Focus:

Machine Design, Automation Equipment Design, Mechanical Engineering, and Manufacturing Systems

EDUCATION

Bachelor of Science in Mechanical Engineering - Manuel S. Enverga University Foundation, Lucena City, Philippines | 2022
Bachelor of Science in Mechatronics - Batangas State University JPLPC, Malvar Campus, Philippines | 2013

LANGUAGES

English - Professional proficiency | Filipino - Native proficiency

PROFESSIONAL STRENGTHS

Engineering Leadership | Strategic Problem Solving | Cross-functional Team Leadership | Project & Stakeholder Management | Customer-focused Engineering | Continuous Improvement | Adaptability | Mentoring & Coaching

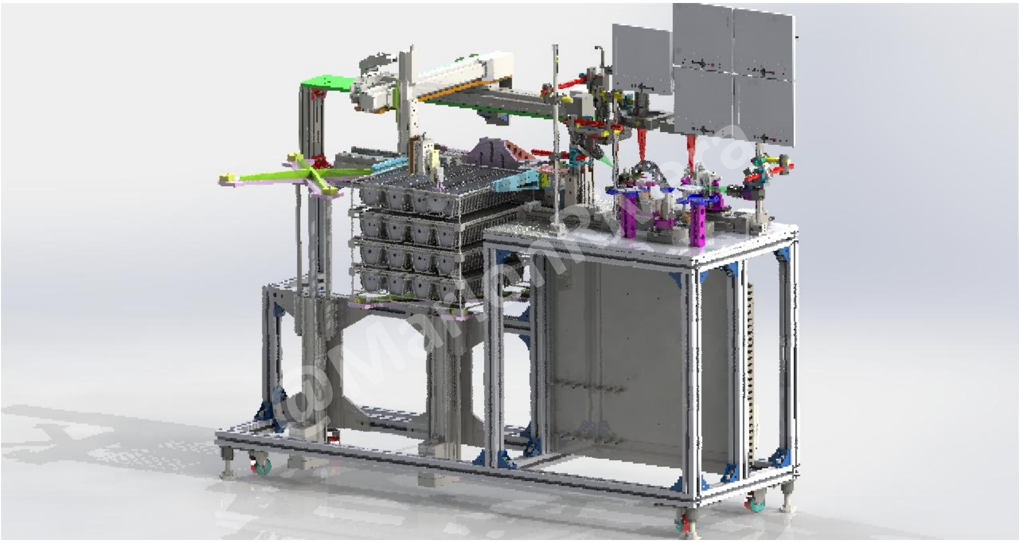
LinkedIn Profile: www.linkedin.com/in/marlon-rivera-49220612a
Portfolio Link: https://marlonrivera-automationengineer.github.io/

ENGINEERING PROJECT PORTFOLIO

[Marlon A. Rivera]
Senior Automation Design & Controls Engineer
Graduate of Bachelor of Science in Mechatronics
Graduate of Bachelor of Science in Mechanical Engineering
https://marlonrivera-automationengineer.github.io/

- an **Automation Design/Programmer Engineer**. With more than **13 years of international experience** in industrial automation, machine design, robotics integration, PLC/HMI programming, and Smart Manufacturing.
- Throughout my career, I have successfully designed, programmed, commissioned, and supported automated manufacturing systems across the **Philippines, Thailand, Vietnam, Korea, and Cambodia**. My experience covers the complete project lifecycle—from concept development, machine design, PLC programming, electrical integration, commissioning, production validation, and continuous improvement.
- Created more than 100 types design of machines, jigs, and fixtures. Attached some of the big projects below.

AUTOMATIC VISUAL MACHINE INSPECTION (AVMI)| Philippines | 2014



Project Overview
- Designed and developed a fully automated Automatic Visual Machine Inspection (AVMI) system to replace manual visual inspection during the manufacturing process of Apple products. The objective of the project was to improve inspection consistency, eliminate human error, increase production efficiency, and enhance product traceability.
- The system integrated industrial sensors, precision positioning, vision inspection technology, and PLC-controlled automation to perform repeatable quality inspections with minimal operator intervention.

Primary Responsibilities
- Machine Concept Development | Mechanical Design | PLC Programming | HMI Development | Vision System Integration | Sensor Integration | Electrical Design Support

Technologies Used
- Automation (Keyence PLC, HMI, Visual Inspection)
- Mechanical (Solidworks 3D and 2D)
- Manufacturing (Industrial Sensors, Camera Inspection, Precision Positioning)

Challenge
- Manual visual inspection depended heavily on operator experience, resulting in inconsistent inspection quality, reduced production efficiency, and the potential for human error during high-volume manufacturing.
- The challenge was to develop an automated inspection system capable of delivering consistent, repeatable, and reliable inspection results while maintaining production cycle time.

Solutions
- Designed and implemented a PLC-controlled automatic inspection system integrating industrial sensors, machine vision technology, precision positioning mechanisms, and automated material handling.
- The inspection process was fully synchronized with the production cycle, allowing automatic product detection, inspection execution, pass/fail decision making, and production data monitoring.

Lessons Learned
- This project strengthened my expertise in automated inspection systems, quality assurance automation, machine vision integration, and production process optimization. It also demonstrated how automation can significantly improve manufacturing quality by replacing manual inspection with reliable and repeatable automated systems.

Auto Laser Welding System | Philippines | 2015

Project Overview

- Designed and developed a fully automated laser welding system for high-volume Apple product manufacturing. The project involved complete machine development from concept design through production commissioning, integrating precision laser welding technology with industrial automation to improve product quality, manufacturing consistency, and production throughput.
- The automation system was designed to support approximately **50,000 units per production line per month**, significantly reducing manual intervention while maintaining stable production performance and high welding precision.

Primary Responsibilities

- Machine Concept Development | Mechanical Design | PLC Programming | Servo Motion Integration | Pneumatic System Design | Machine Assembly | Production Support

Technologies Used

- Automation (Mitsubishi PLC, 3 Axis IAI Robot)
- Mechanical (Solidworks 3D and 2D)
- Manufacturing (YAG Precision Laser Welding, SMC Pneumatic System)

Challenge

- The project required maintaining consistent laser welding quality during continuous high-volume production while minimizing operator intervention and ensuring repeatable positioning accuracy.
- The machine also needed to maintain stable cycle time and welding precision under continuous manufacturing conditions.

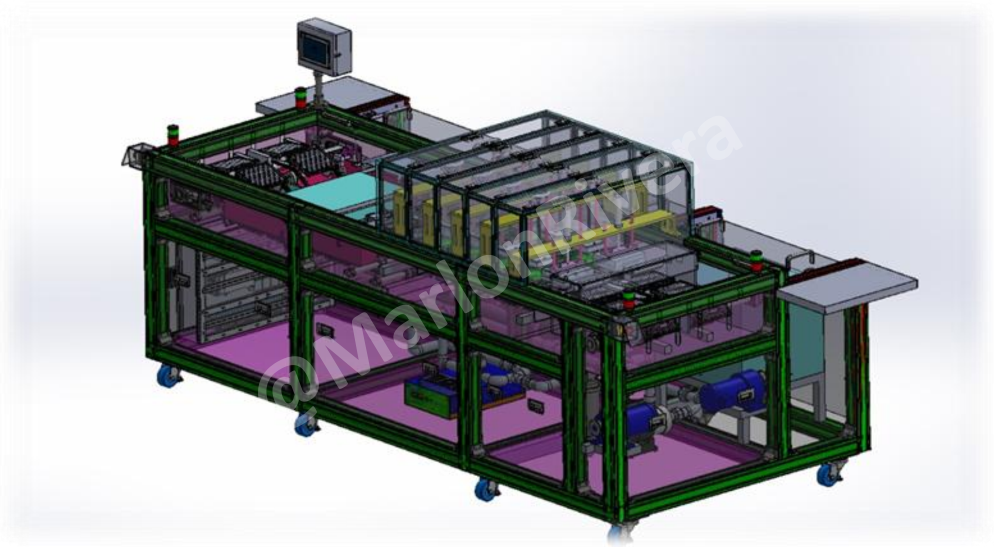
Solutions

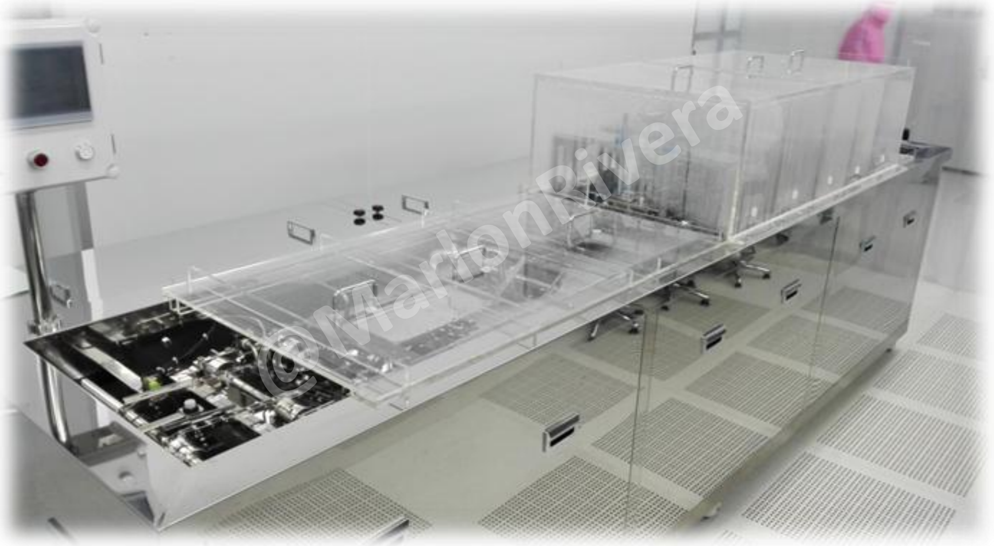
- Developed a fully automated laser welding system integrating PLC control, precision motion control, servo positioning, pneumatic automation, and HMI operation.
- The control sequence synchronized product positioning, laser activation, safety interlocks, and machine diagnostics to ensure repeatable production quality while reducing manual handling.
- Machine validation included production testing, process optimization, and operator training before production handover.

Lessons Learned

- This project strengthened my experience in precision manufacturing, machine integration, production commissioning, and large-scale industrial automation. It also reinforced the importance of designing reliable automation systems that combine mechanical engineering, controls engineering, and manufacturing process optimization.

AUTO SPRAY MACHINE | Philippines/Vietnam | 2015





Project Overview
- Led the complete engineering design and automation development of a fully automatic spray machine for Samsung Camera manufacturing. The project was developed to automate the spraying process, improve product quality, increase production consistency, and reduce manual operation.
- Following the successful implementation in the Philippines, the system was upgraded to Version 2 and deployed internationally to IM Vietnam.

Primary Responsibilities
- Customer Requirement Analysis | Machine Concept Development | Mechanical Design | PLC Programming | HMI Development | Pneumatic System Design | Customer Acceptance Support

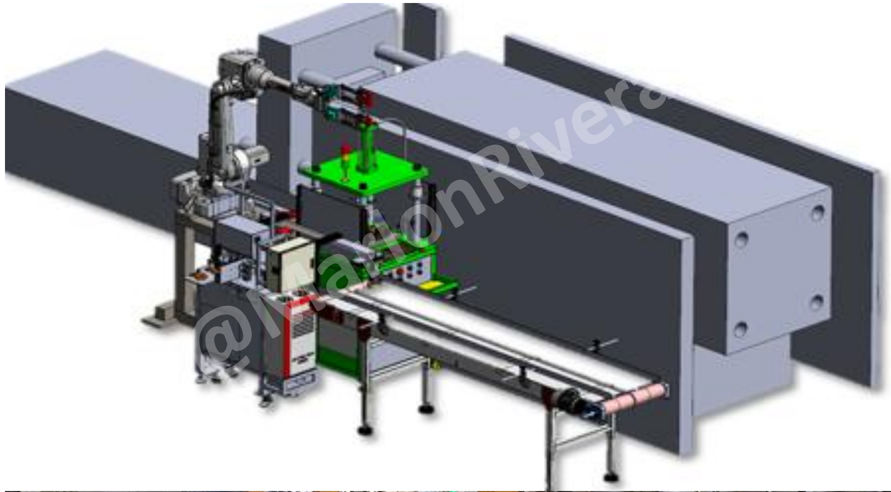
Technologies Used
- Automation (LSIS PLC, HMI, DI Pump)
- Mechanical (Solidworks 3D and 2D)
- Manufacturing (Pneumatic Systems, Precision Automation, Production Equipment)

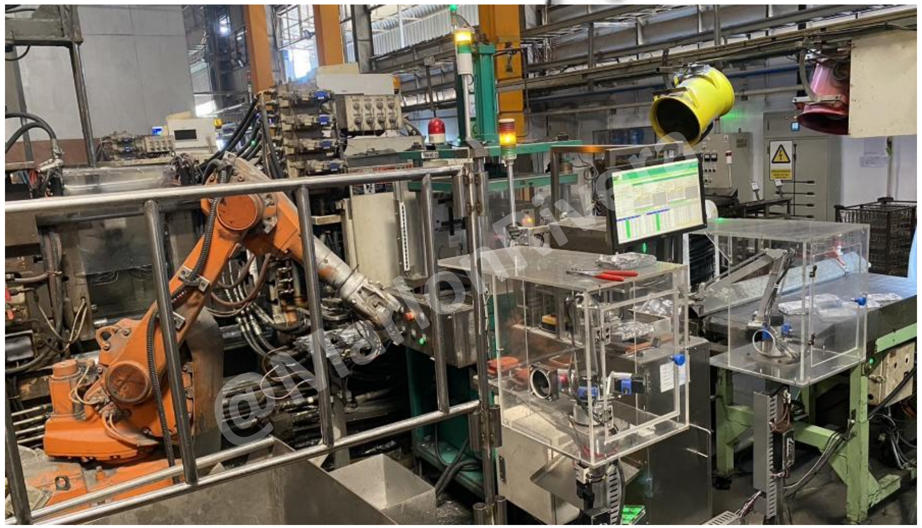
Challenge
- Samsung Camera production required a highly consistent automatic spraying process capable of maintaining stable coating quality while minimizing operator dependency.
- The machine needed to achieve repeatable production quality, improve manufacturing efficiency, and remain reliable during continuous operation.

Solutions
- Developed a fully automated spraying system integrating PLC control, servo positioning, pneumatic automation, and precision mechanical design.
- The automation sequence synchronized product handling, positioning, spraying operations, and safety interlocks while allowing operators to monitor production through an HMI interface.

Project Impact
- The Auto Spray Machine became one of the most significant engineering projects in my career.
- Following successful implementation in the Philippines, the system was further enhanced into Version 2 and deployed to IM Vietnam, demonstrating the scalability and reliability of the original engineering design.
- The project was also recognized through a Korean Government patent, highlighting the innovation and engineering value delivered through the solution.
- The successful completion of this project contributed to my recognition as Employee of the Year (2016) for outstanding engineering performance, innovation, leadership, and project execution.

FULLY AUTOMATED DIECASTING PRODUCTION LINE | Cambodia| 2020





Integrated Production Line
- The fully automated diecasting production line was designed as an integrated manufacturing system rather than a collection of individual machines.
- The automation concept connected multiple production processes into a coordinated manufacturing flow that improved productivity, reduced manual handling, and supported future digital manufacturing initiatives.

Project Overview
- Led the engineering design and implementation of a fully automated diecasting production line to improve manufacturing efficiency, product quality, equipment utilization, and production consistency.
- The project focused on transforming conventional diecasting operations into an integrated automated production line by combining industrial robots, automatic transfer systems, trimming automation, deburring automation, production monitoring, and Smart Manufacturing concepts.
- The objective was to reduce manual handling, improve process stability, enhance operator safety, and support future manufacturing expansion through scalable automation.

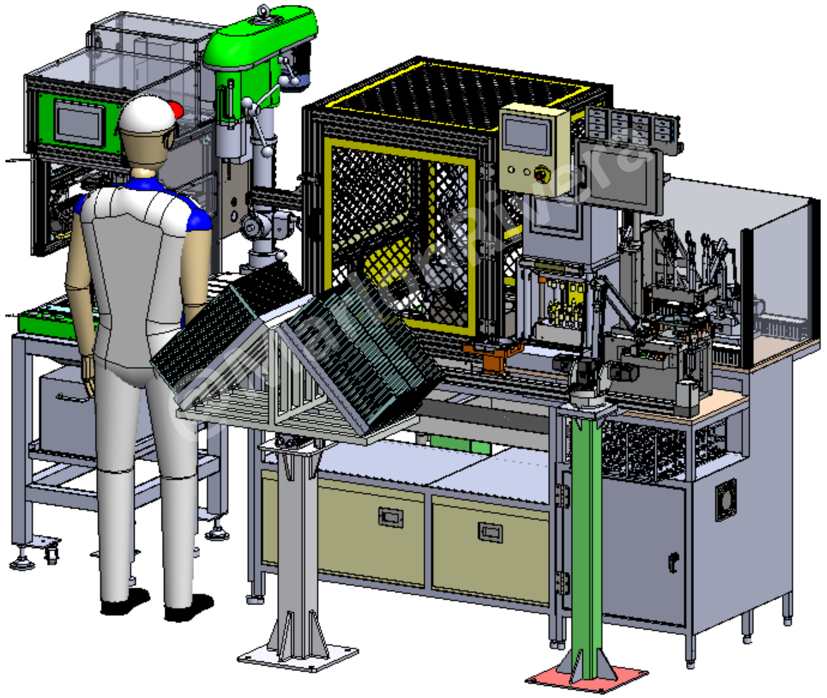
Primary Responsibilities
- Production Line Automation Planning | Machine Concept Development | Mechanical Design | PLC Programming | Robot Integration | Transfer System Design | Project Leadership

Technologies Used
- Automation (PLC, HMI, Industrial Networking, Servo Motion)
- Mechanical (Solidworks 3D and 2D)
- Robotics (ABB Robot, LCPL Robot, Automatic Transfer Systems)
- Manufacturing (Smart Manufacturing, Industry 4.0, OEE Monitoring, Production Monitoring)

Challenge
- The existing production process relied heavily on manual handling between manufacturing operations, creating opportunities for inconsistent cycle times, increased operator workload, and limited production scalability.
- The challenge was to integrate multiple production processes into one automated manufacturing line while maintaining stable production flow, reliable equipment communication, and safe operation.
- The solution also needed to support future Smart Manufacturing initiatives through centralized production monitoring and machine data collection.

Solutions
- Developed an integrated automation concept combining industrial robots, automated transfer systems, trimming automation, deburring automation, production monitoring, and PLC-controlled machine coordination..
- The production line was designed to synchronize multiple manufacturing processes through centralized automation, reducing manual intervention while improving equipment utilization and manufacturing efficiency.
- The project also established the foundation for Smart Manufacturing by integrating production monitoring, OEE data collection, and real-time machine status visualization.

ROBOTIC DEBURRING AUTOMATION CELL | Thailand | 2022



Project Overview

- Designed and implemented a robotic deburring automation cell to replace manual deburring operations and improve manufacturing consistency, operator safety, and production efficiency.
- The project integrated industrial robotics, PLC control, servo motion, pneumatic systems, and automated workpiece handling into a reliable production cell capable of operating continuously in a manufacturing environment.
- The objective was to reduce manual labor, improve product quality, minimize operator fatigue, and achieve repeatable deburring results.

Primary Responsibilities

- Robot Cell Layout Design | Machine Concept Development | Robot Integration | PLC Programming | HMI Development | Servo Motion Integration| Safety System Integration

Technologies Used

- Automation (Panasonic PLC, HMI, Servo Motor, Nachi 6 axis Robot)
- Mechanical (Solidworks 3D and 2D)
- Manufacturing (Pneumatic Systems, Robot End-of-Arm Tooling (EOAT), Production Equipment)

Challenge

- Manual deburring created inconsistent product quality, variable cycle times, and increased operator workload. The process also required precise tool positioning while maintaining repeatable quality over continuous production.
- The engineering challenge was to automate the deburring process while ensuring robot accuracy, safe operation, and reliable communication between the robot, PLC, sensors, and production equipment.

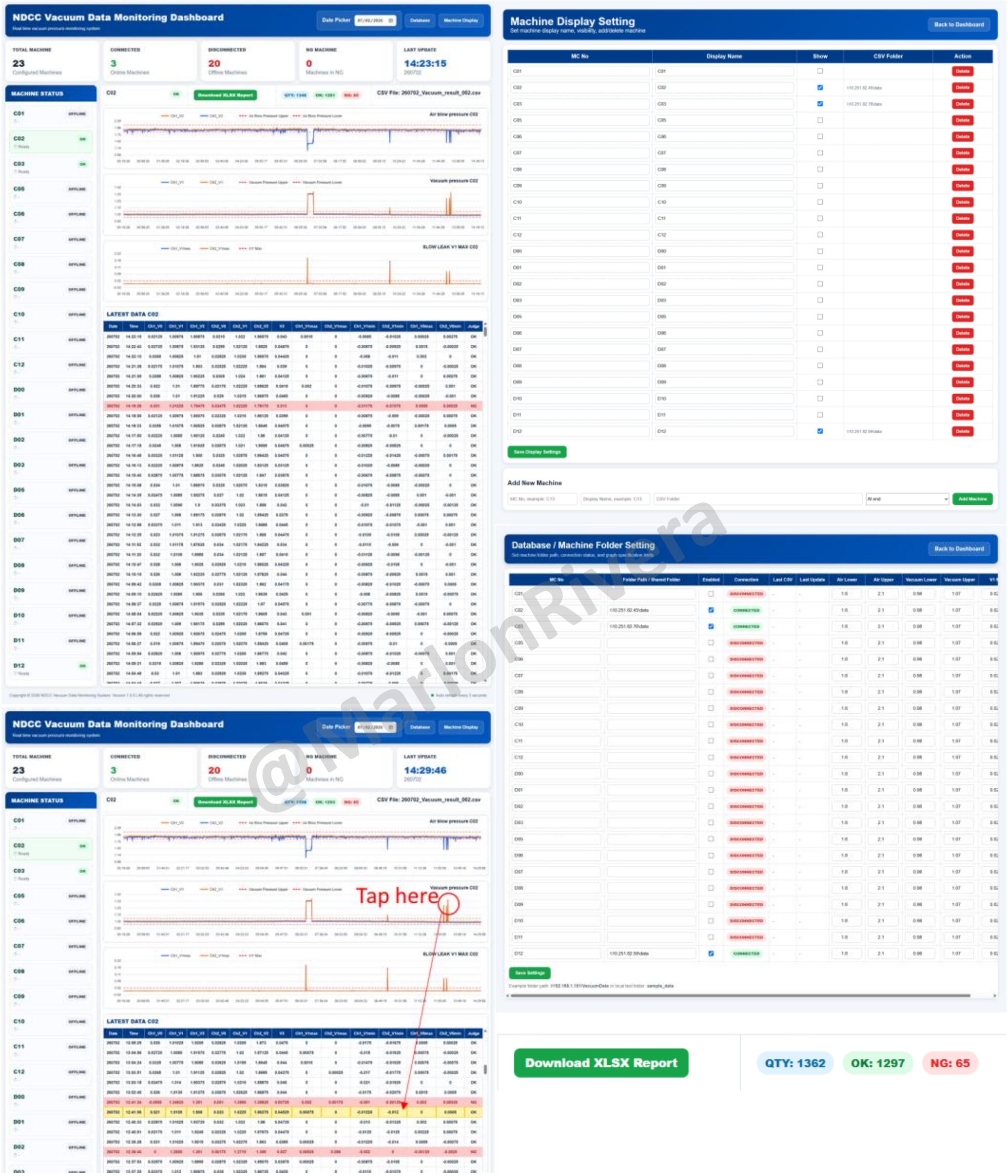
Solutions

- Developed a fully integrated robotic deburring cell combining industrial robotics, PLC control, servo positioning, pneumatic clamping, safety interlocks, and automated part handling.
- The automation sequence synchronized robot movement, fixture positioning, sensor verification, and machine communication to provide repeatable deburring performance while reducing manual intervention.

Project Impact

- Reduced manual deburring operations, Improved product consistency, Increased production efficiency, Enhanced operator safety, Improved process repeatability, Stable robotic production

VACUUM MONITORING | Cambodia/Thailand | 2025



Project Overview

- Designed and developed a centralized Vacuum Monitoring and Predictive Maintenance System for manufacturing equipment to provide real-time monitoring of vacuum pressure, machine health, historical trends, and alarm conditions.
- The system automatically collects vacuum data from multiple production machines, stores historical records, generates real-time dashboards, and provides engineering teams with early visibility of abnormal machine conditions.

Primary Responsibilities

- System Architecture Design | Vacuum Data Collection | Python Software Development | Database Development | Dashboard Development | Alarm Management | Historical Trend Analysis | User Management | Machine Configuration | System Deployment

Technologies Used

- Industrial (Vacuum Sensors, Industrial Data Collection, Machine Monitoring)
- Software (Python, FastAPI, HTML, JavaScript, SQL Database)
- Manufacturing (Predictive Maintenance, Historical Trending, Alarm Management, Eqpt. Monitoring)

Challenge

- Vacuum performance is critical to manufacturing quality and machine reliability. Previously, engineers relied on manual observation and machine-side indicators, making it difficult to identify gradual performance degradation or investigate historical issue.
- The challenge was to create a centralized monitoring platform capable of collecting real-time vacuum data from multiple machines while providing historical analysis, alarm management, and intuitive dashboards for maintenance personnel.

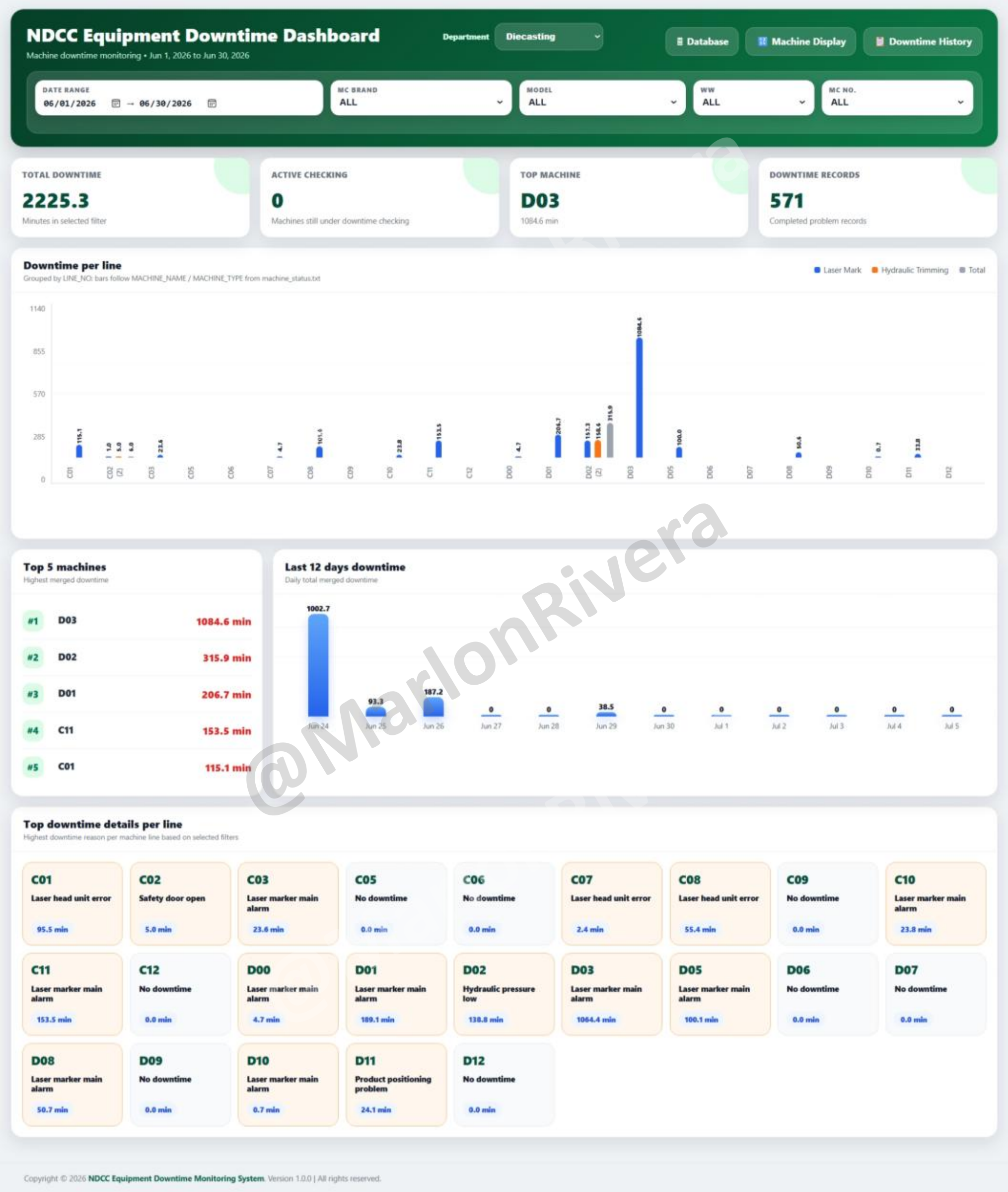
Solutions

- Developed a centralized web-based monitoring platform using Python and FastAPI to automatically collect vacuum data, maintain historical records, generate real-time graphical dashboards, and display machine health information.
- The system supports configurable machine settings, alarm logging, historical trend visualization, and centralized monitoring for multiple production machines, allowing maintenance teams to respond quickly to abnormal conditions.

Project Impact

- Real-time vacuum monitoring✓ Centralized machine dashboard✓ Historical trend recording✓ Alarm event logging✓ Improved preventive maintenance✓ Faster fault investigation✓ Enhanced equipment reliability

SMART FACTORY & OEE MONITORING SYSTEM | Cambodia | 2026



Project Overview

- Led the design and development of a Smart Factory Monitoring and Overall Equipment Effectiveness (OEE) system to provide real-time production visibility across manufacturing operations.
- The system centralized machine status, downtime monitoring, production data collection, OEE analysis, and production dashboards into a single platform, enabling engineers and management to monitor factory performance in real time.
- The solution was developed to support Industry 4.0 initiatives by integrating industrial automation with software development and manufacturing analytics.

Primary Responsibilities

- Smart Factory System Architecture | OEE System Design | PLC Data Integration | Production Data Collection | Python Software Development | Dashboard Development | Database Design

Technologies Used

- Automation (Mitsubishi PLC, Machine Status Readers, Industrial Networking)
- Software (Python, FastAPI, HTML, JavaScript, SQL, REST API)
- Manufacturing (OEE Monitoring, Downtime Analysis, Production Dashboard, Smart Manufacturing)

Challenge

- Manufacturing data was previously distributed across multiple production areas with limited real-time visibility. Engineers and management required immediate access to machine status, downtime information, production performance, and OEE metrics to support faster decision-making and continuous improvement.
- The challenge was to integrate production equipment into a centralized monitoring platform while ensuring reliable communication, accurate data collection, and user-friendly visualization.

Solutions

- Designed and developed a Smart Factory Monitoring System integrating PLC communication, machine data collection, centralized databases, real-time dashboards, and production analytics.
- The platform automatically collected machine status, downtime events, production information, and OEE data while presenting the information through web-based dashboards accessible to engineering and management teams.

Project Impact

- Centralized production monitoring✓ Real-time machine status visualization✓ OEE performance tracking✓ Downtime history recording✓ Faster engineering response✓ Improved production visibility✓ Foundation for Industry 4.0 implementation